

SQL Business Problem Solving

Retail Sales Analysis: Proof of Concept (POC)

Data Analytics Consultant

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Analysis Objectives & Scope



Sales Operations

Retrieval and filtering of specific transactional records for operational auditing.



Customer Profiling

Analyzing demographics such as age, gender, and buying patterns for segmentation.



Advanced Trends

Time-series analysis and performance ranking using CTEs and Window Functions.

| 01. Daily Operational Audit

Business Context

During financial reconciliation, auditors often need to pull raw transaction data for a specific date to verify point-of-sale logs against bank deposits.

Insight: Enables localized spot-checks of data integrity on specific high-traffic dates.

```
SELECT * FROM retail_sales WHERE sale_date =  
'2022-11-05';
```

Key Concepts

Basic Filtering, Date Literal Formatting

02. Campaign Performance

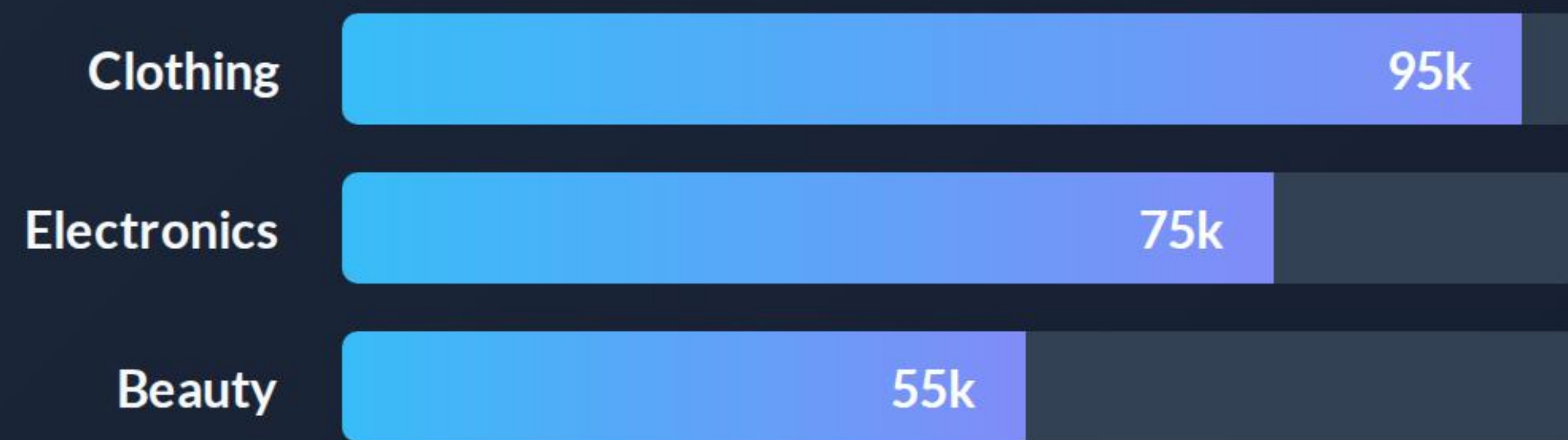
Use Case: Evaluate the success of 'November Clothing' bulk-buy promotions.

```
SELECT * FROM retail_sales WHERE category = 'Clothing'  
AND quantity ≥ 4 AND MONTH(sale_date) = 11;
```

Insight: Identifies bulk purchasers during seasonal peaks, aiding in inventory replenishment strategy.



03. Category Revenue Distribution



```
SELECT category, SUM(total_sale) FROM  
retail_sales GROUP BY category;
```

Business Insight

Quantifying the primary revenue drivers. Clothing consistently leads in net volume, suggesting prioritized shelf-space allocation.

04. Beauty Segment Demographics



Average Customer Profile

```
SELECT customer_id, AVG(age) FROM  
retail_sales WHERE category = 'Beauty' GROUP BY  
customer_id;
```

Insight: Understanding the age demographic of beauty customers allows for hyper-targeted social media marketing campaigns aimed at the identified average age bracket.

05. Identifying "Whale" Transactions

> 1000

Threshold for Premium Orders

```
SELECT * FROM retail_sales WHERE total_sale  
> 1000;
```

Business Value

Flags high-ticket sales for potential fraud review or specialized customer service follow-ups to encourage repeat loyalty in the luxury segment.

06. Behavioral Segmentation Matrix

Category	Gender	Transaction Count	Status
Clothing	Female	452	High Engagement
Beauty	Male	128	Growth Potential
Electronics	Male	389	Primary Segment

```
SELECT category, gender, COUNT(transactions_id) FROM retail_sales GROUP BY gender, category ORDER BY 1;
```


07. Seasonal Peak Analysis

Performance Trends

Using Common Table Expressions (CTEs) to isolate the highest-performing month in each fiscal year.

Insight: Seasonality peaks often align with holiday cycles. This query provides a scientific basis for seasonal staffing and inventory buildup.

```
WITH cte AS ( SELECT YEAR(sale_date) yr, MONTH(sale_date)
mon, ROUND(AVG(total_sale), 2) avgSales FROM retail_sales GROUP
BY YEAR(sale_date), MONTH(sale_date) ) SELECT yr, MAX(avgSales)
FROM cte GROUP BY yr;
```


08. VIP Loyalty Identification

CUST-1042

Revenue: \$5,420

CUST-2189

Revenue: \$4,910

CUST-0988

Revenue: \$4,200

```
SELECT TOP 5 customer_id, SUM(total_sale)
FROM retail_sales GROUP BY customer_id ORDER BY 2
DESC;
```

Business Use Case

Identifies the "Top 5" customers by lifetime value. These individuals are prioritized for exclusive loyalty rewards and white-glove service programs.

09. Customer Penetration Analysis



Market Reach Calculation

```
SELECT category, COUNT(DISTINCT customer_id)
FROM retail_sales GROUP BY category;
```

Insight: Distinguishes between categories with high-frequency repeat buyers vs. categories with wide reach but low frequency.

| 10. Staffing & Shift Optimization

Time-of-Day Dynamics

Categorizing transactions into Morning, Afternoon, and Evening shifts based on sale time timestamps.

Business Insight: If "Shift 2" (Afternoon) shows 60% of orders, staffing levels should be increased between 12:00 and 17:00 to reduce checkout friction.

```
WITH cte AS ( SELECT *, CASE WHEN DATEPART(HOUR,
sale_time) < 12 THEN 'Morning' WHEN DATEPART(HOUR, sale_time)
BETWEEN 12 AND 17 THEN 'Afternoon' ELSE 'Evening' END AS Shift
FROM retail_sales ) SELECT Shift, COUNT(*) FROM cte GROUP BY
Shift;
```


| Key Technical Competencies

Logic

Advanced conditional logic using **CASE Statements** and multi-step processing with **CTEs**.

Aggregation

Complex **SUM/AVG** calculations paired with **Multi-level Grouping** and filtering.

Temporal

Sophisticated **Date Functions** including Month/Year extraction and **DATEPART** hour analysis.

Thank You

Questions & Discussion

Deliverable: SQL Server Portfolio - Retail POC

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